

AMEBIASIS CASE STUDY

1. Case Introduction

Amebiasis is a parasitic infection of the human gastrointestinal tract caused by the protozoan *Entamoeba histolytica*. It is a significant global health problem, particularly in developing countries with poor sanitation, where it contributes to substantial morbidity and mortality. The infection primarily affects the large intestine, leading to a spectrum of clinical manifestations ranging from asymptomatic colonization to severe dysentery and life-threatening extra-intestinal complications, most notably amoebic liver abscess. Understanding its epidemiology, clinical features, and management is crucial for effective control and treatment. (World Health Organization, 1997; CDC, 2020)

2. Definition

Amebiasis is defined as an infection with the protozoan parasite *Entamoeba histolytica*, irrespective of whether the individual is symptomatic or asymptomatic. Clinical disease occurs when the trophozoite form of the parasite invades the intestinal mucosa, causing tissue destruction and inflammation, primarily manifesting as amoebic dysentery or amoebic liver abscess. (Stanley, 2003; Haque et al., 2003)

3. Epidemiological Study

Global and National Burden:

- a. Global: Amebiasis is the second leading cause of death from a parasitic disease worldwide. It is estimated that *E. histolytica* infects approximately 50 million people annually, resulting in up to 100,000 deaths. The disease is endemic in regions with poor socioeconomic conditions, including parts of the Indian subcontinent, Africa, Southeast Asia, and Central and South America. (WHO, 1997; CDC, 2020)
- b. National Situation (Bangladesh): In Bangladesh, amebiasis is common because of overcrowding, poor sanitation, unsafe drinking water, and inadequate hygiene practices. Both children and adults are affected, especially in rural and slum areas. (Haque et al., 2003; ICDDR,B, 2018)

Mode of Transmission:

- a. Fecal-Oral Route: The primary mode of transmission is through the ingestion of mature cysts from fecally contaminated food or water. (CDC, 2020)
- b. Direct Contact: Can occur via direct fecal-oral contact, particularly in settings with poor personal hygiene or during certain sexual practices (oral-anal contact). (Stanley, 2003)
- c. Vectors: Mechanical vectors like flies and cockroaches can transfer cysts from feces to uncovered food. (WHO, 1997)

Risk Factors:

- a. Poor sanitation and lack of safe drinking water. (WHO, 1997)
- b. Low socioeconomic status and overcrowding. (Haque et al., 2003)
- c. Travel to or residence in endemic areas. (CDC, 2020)
- d. Institutional living (e.g., prisons, mental health facilities). (Stanley, 2003)
- e. Immunocompromised states (e.g., HIV/AIDS, malnutrition). (WHO, 1997)
- f. Men who have sex with men (MSM). (CDC, 2020)

Clinical Features: Intestinal Amebiasis:

- a. Asymptomatic Cyst Passer: The most common presentation (90% of cases). (Stanley, 2003)
- b. Acute Amoebic Dysentery: Gradual onset of abdominal pain, tenesmus, and frequent, small-volume diarrhea containing blood and mucus (classic "anchovy paste" or "raspberry jelly" stool). Fever is usually absent or mild. (Haque et al., 2003)
- c. Fulminant Colitis: Severe form with profuse bloody diarrhea, high fever, diffuse abdominal pain, and risk of toxic megacolon or bowel perforation. (CDC, 2020)
- d. Chronic Amebiasis: Mimics inflammatory bowel disease with intermittent diarrhea, weight loss, and abdominal pain. (Stanley, 2003)
- e. Ameboma: A localized, inflammatory mass in the colon wall (usually cecum) that can mimic carcinoma. (Bennett et al., 2019)

Extra-intestinal Amebiasis:

Amoebic Liver Abscess (ALA): The most common extra-intestinal manifestation. Presents with high-grade fever, right upper quadrant pain radiating to the shoulder, and tender hepatomegaly. Jaundice is rare. Diarrhea is present in only a minority of patients.

Other rare sites include the pleuropulmonary system, brain, and skin. (Stanley, 2003; Haque et al., 2003)

4. Patient Profile

Particular	Patient Information
Name	Mrs. Nurjahan Begum
Age	32 years
Sex	Female
Religion	Muslim
Marital Status	Married
Occupation	Garment worker
Address	Chattogram, Bangladesh
Educational Status	SSC
Family Members	4

Socioeconomic Status	Middle-income family
Date of Admission/Consultation	June 28, 2026
Medical Diagnosis	Intestinal Amebiasis

5. Chief Complaint

The patient complained of:

- a. Frequent loose stools for 5 days.
- b. Abdominal cramps and lower abdominal pain for 4 days.
- c. Passage of mucus with stool for 3 days.
- d. Mild weakness and loss of appetite.
- e. Occasional low-grade fever.

Previous Medical History

The patient had no history of diabetes mellitus, hypertension, tuberculosis, or other major chronic diseases. She reported occasional episodes of diarrhea in the past but had never been diagnosed with amebiasis. There was no history of recent hospitalization or surgery. She had not taken any regular medication before the present illness.

Family History

There was no significant family history of chronic gastrointestinal disease. However, one family member had experienced mild diarrhea approximately two weeks earlier. There was no known history of tuberculosis, diabetes, hypertension, or hereditary disease in the family.

6. Diagnostic Investigation

Investigation	Normal Range	Patient's Result	Clinical Significance
Hemoglobin (Hb)	Female: 12–16 g/dL	11.8 g/dL	Slightly decreased; may be associated with nutritional deficiency or blood loss due to intestinal infection.
Total WBC count	4,000–11,000 cells/ μ L	12,400 cells/ μ L	Mild leukocytosis, suggesting an ongoing infection/inflammatory response.
Neutrophils	40–70%	74%	Mildly increased, indicating an acute inflammatory/infective response.
Platelet count	150,000–450,000/ μ L	285,000/ μ L	Within normal range.
ESR	Female: 0–20 mm/hr	28 mm/hr	Mildly elevated, suggesting inflammation.

Serum Sodium	135–145 mmol/L	136 mmol/L	Within normal range; no significant sodium imbalance.
Serum Potassium	3.5–5.0 mmol/L	3.7 mmol/L	Within normal range.
Stool examination – RBC	Absent	Present	Suggests intestinal mucosal damage and dysentery.
Stool examination – WBC/pus cells	Absent or few	Increased	Indicates intestinal inflammation.
Stool examination – <i>Entamoeba histolytica</i>	Not detected	Detected	Supports the diagnosis of intestinal amebiasis.
Stool occult blood	Negative	Positive	Indicates blood loss from the intestinal mucosa.

Patient's Result

The patient's investigations showed mild leukocytosis, increased neutrophils, elevated ESR, slightly decreased hemoglobin, and abnormal stool findings. Stool examination detected *Entamoeba histolytica*, with RBCs, increased pus cells, and positive occult blood.

Clinical Significance

The laboratory findings indicate an active intestinal infection with inflammation and mucosal injury. Detection of *Entamoeba histolytica* in the stool supports the diagnosis of intestinal amebiasis (amoebic dysentery). The presence of RBCs and occult blood

suggests intestinal mucosal damage, while increased WBCs and ESR indicate an inflammatory response.

7. Medical Diagnosis + Differential Diagnosis

Medical Diagnosis

- a. Stool Examination (O&P): Microscopy for trophozoites (in liquid stool) or cysts (in formed stool). Cannot differentiate *E. histolytica* from non-pathogenic *E. dispar*. (CDC, 2020)
- b. Stool Antigen Detection: ELISA-based tests specific for *E. histolytica* adhesin. Preferred method for diagnosing active intestinal infection. (Haque et al., 2003)
- c. Serology: Highly sensitive for extra-intestinal amebiasis (especially ALA), where stool tests are often negative. A positive IgG test indicates past or current infection. (Stanley, 2003)
- d. Molecular Methods (PCR): Highly sensitive and specific for detecting *E. histolytica* DNA in stool or abscess fluid. (Bennett et al., 2019)
- e. Imaging (for ALA): Ultrasound or CT scan reveals a single, right-lobe, hypoechoic lesion without a prominent wall. (WHO, 1997)
- f. Diagnostic Aspiration: Aspirate from an ALA is classically an "anchovy paste"-like fluid (necrotic tissue, not pus), and trophozoites are found in only the wall of the abscess, not the center. (Stanley, 2003)

Differential Diagnosis

- a. For Intestinal Amebiasis: Bacterial dysentery (shigellosis, campylobacteriosis), Inflammatory Bowel Disease (Ulcerative Colitis, Crohn's disease), Colorectal carcinoma, Ischemic colitis. (CDC, 2020; Bennett et al., 2019)
- b. For Amoebic Liver Abscess: Pyogenic liver abscess, Hepatocellular carcinoma, Echinococcal (hydatid) cyst, Cholecystitis. (Stanley, 2003; Haque et al., 2003)

8. Medical Management, Pharmacological Management and Drug Adverse Effects

Medical Management

- a. Maintain hydration and electrolyte balance. (WHO, 1997)
- b. Nutritional support. (CDC, 2020)
- c. Monitor stool frequency and abdominal pain. (Stanley, 2003)
- d. Maintain personal hygiene. (WHO, 1997)
- e. Isolation precautions if necessary. (CDC, 2020)

Pharmacological Management

- a. Tissue Amebicides (for trophozoites):
 - i. Tinidazole: 2g once daily orally for 3–5 days (preferred due to fewer side effects and shorter course). (Bennett et al., 2019)
 - ii. Metronidazole: 500–750 mg orally or IV three times daily for 7–10 days. (Stanley, 2003)
- b. Luminal Amebicides (for cysts in the gut lumen): Must follow the tissue agent to eradicate colonization and prevent recurrence.
 - i. Paromomycin: 25–35 mg/kg/day orally in 3 divided doses for 7 days. (CDC, 2020)
 - ii. Diloxanide Furoate: 500 mg orally three times daily for 10 days. (WHO, 1997)
- c. For ALA: Treatment is primarily medical with a tissue amebicide (Metronidazole or Tinidazole), followed by a luminal agent. Therapeutic aspiration is reserved for large abscesses (>10 cm), left-lobe abscesses, impending rupture, or failure to respond to drugs in 5–7 days. (Stanley, 2003; Haque et al., 2003)

Drug Adverse Effects

- a. Metronidazole: Metallic taste, nausea, anorexia, dark urine, peripheral neuropathy (prolonged use), disulfiram-like reaction with alcohol. (Bennett et al., 2019)
- b. Tinidazole: Bitter/metallic taste, nausea, dizziness, headache. (CDC, 2020)
- c. Paromomycin: Nausea, abdominal cramps, diarrhea. Contraindicated in intestinal obstruction. (WHO, 1997)
- d. Diloxanide Furoate: Flatulence, nausea, pruritus, urticaria. (Stanley, 2003)

9. Nursing Care Plan

Assessment	Nursing Diagnosis	Planning	Nursing Interventions	Rationale	Evaluation
Frequent loose stools with mucus for 5 days.	Diarrhea related to intestinal infection as evidenced by frequent loose stools and mucus.	Patient will have reduced diarrhea and improved stool consistency.	1. Monitor stool frequency and characteristics. 2. Encourage ORS and fluids. 3. Maintain perineal hygiene. 4. Administer prescribed medication.	1. Assesses severity of diarrhea. 2. Prevents dehydration. 3. Prevents skin irritation. 4. Treats the underlying infection.	Stool frequency decreased and consistency improved.

Lower abdominal pain and cramps for 4 days.	Acute pain related to intestinal inflammation as evidenced by abdominal pain and cramps.	Patient will report reduced pain and improved comfort.	1. Assess pain intensity. 2. Provide comfortable position and rest. 3. Give prescribed medication. 4. Reassess pain.	1. Determine severity of pain. 2. Promotes comfort. 3. Relieves symptoms and treats the cause. 4. Evaluates effectiveness of treatment.	Patient reported decreased abdominal pain and improved comfort.
Weakness and loss of appetite; Hb 11.8 g/dL.	Imbalanced nutrition: less than body requirements related to decreased appetite and gastrointestinal disturbance	Patient will improve food intake and maintain adequate nutrition.	1. Assess appetite and food intake. 2. Give small, frequent, easily digestible meals. 3. Encourage adequate fluids. 4.	1. Identifies nutritional problems. 2. Improves food tolerance. 3. Prevents dehydration. 4. Monitors	Patient tolerated meals and appetite improved.

	e.		Monitor weight if required.	nutritional status.	
Frequent diarrhea with risk of fluid loss.	Risk for inadequate fluid volume related to excessive fluid loss through diarrhea.	Patient will maintain adequate hydration and stable vital signs.	1. Monitor intake and output. 2. Assess for dehydration. 3. Encourage ORS and fluids. 4. Monitor vital signs.	1. Determines fluid balance. 2. Detects dehydration early. 3. Replaces fluid and electrolyte losses. 4. Detects physiological changes.	Patient remained adequately hydrated with stable vital signs.
Patient has insufficient knowledge about prevention of	Deficient knowledge related to lack of information about	Patient will verbalize understanding of prevention and	1. Explain transmission of amebiasis. 2. Teach proper	1. Improves disease awareness. 2. Prevents	Patient correctly explained preventive and treatment

amebiasis.	disease prevention and treatment.	treatment.	handwashing. 3. Advise safe food and water practices. 4. Explain importance of completing medication.	fecal-oral transmission. 3. Prevents reinfection. 4. Promotes effective treatment.	measures.
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10. Health Education

- a. Safe Water: Drink boiled, filtered, or bottled water. Avoid ice made from tap water in endemic areas. (WHO, 1997)
- b. Food Hygiene: Wash all fruits and vegetables thoroughly, preferably peeling them yourself. Avoid street vendor food and undercooked items. "Cook it, peel it, or forget it." (CDC, 2020)
- c. Hand Hygiene: Wash hands thoroughly with soap and water after using the toilet and before handling, preparing, or eating food. (WHO, 1997)
- d. Sanitation: Promote the use of sanitary latrines and prevent open defecation. (Haque et al., 2003)
- e. Prevention of Spread: Infected individuals should avoid preparing food for others until treatment is completed and stool tests are negative. Safe sexual practices should be encouraged. (CDC, 2020)

11. Disease Prognosis

With prompt and appropriate treatment, the prognosis for uncomplicated intestinal amebiasis and amoebic liver abscess is excellent. Most patients recover completely with a very low mortality rate (<1%). However, the prognosis worsens significantly with complications such as fulminant colitis, toxic megacolon, bowel perforation, or a ruptured liver abscess into the pericardial, pleural, or peritoneal cavities, where mortality rates can be high. The host's immune status and nutritional state are also critical prognostic factors. (Stanley, 2003; Haque et al., 2003)

12. Conclusion

Amebiasis remains a major public health challenge intrinsically linked to poverty, poor sanitation, and unsafe water. The ability of *E. histolytica* to exist as a silent cyst passer complicates control efforts. A high index of suspicion is needed for diagnosis, which hinges on distinguishing the pathogenic *E. histolytica* from the morphologically identical *E. dispar*. Successful treatment requires a two-pronged pharmacological approach: a tissue agent like tinidazole to kill invasive trophozoites, followed by a luminal agent like paromomycin to eradicate intestinal cysts and prevent transmission. Ultimately, prevention through improved sanitation and health education is the cornerstone of long-term disease control. (WHO, 1997; CDC, 2020)

13. References

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HYPERTENSION CASE STUDY

1. Case Introduction

Hypertension is one of the most common non-communicable diseases worldwide and is a major public health concern. It is often called the "silent killer" because many patients remain asymptomatic for years while damage gradually occurs to vital organs such as the heart, brain, kidneys, and eyes. Persistent elevation of blood pressure increases the risk of cardiovascular diseases, stroke, heart failure, and renal failure. Early diagnosis, proper treatment, lifestyle modification, and regular follow-up are essential to control hypertension and prevent complications. (WHO, 2021; CDC, 2022)

2. Definition

Hypertension is a chronic medical condition in which the blood pressure in the arteries remains persistently elevated above the normal level. According to the World Health Organization (WHO), hypertension is generally defined as:

- a. Systolic Blood Pressure (SBP) ≥ 140 mmHg
- b. and/or Diastolic Blood Pressure (DBP) ≥ 90 mmHg measured on at least two separate occasions. (WHO, 2021)

3. Epidemiological Study

Global and National Burden:

Global: An estimated 1.28 billion adults aged 30–79 years have hypertension, with the majority (two-thirds) living in low- and middle-income countries. It is the leading single risk factor for cardiovascular disease and all-cause mortality worldwide. (WHO, 2021)

National Situation (Bangladesh): The Bangladesh Demographic and Health Survey (BDHS) indicates that approximately one in four to one in five adults has hypertension, with a high proportion unaware of their condition. Prevalence is higher in urban than rural areas, but it is rapidly increasing in rural populations as well. (BDHS, 2018; NCD Risk Factor Collaboration, 2020)

Mode of Transmission:

Hypertension is not a communicable disease; therefore, it has no mode of transmission. It develops due to genetic, environmental, and lifestyle-related factors. (WHO, 2021)

Risk Factors:

Non-Modifiable:

- a. Age (risk increases with advancing age) (CDC, 2022)
- b. Family history of hypertension (AHA, 2020)
- c. Gender (higher risk in men before age 65, and in women after menopause) (WHO, 2021)
- d. Race/Ethnicity (AHA, 2020)

Modifiable:

- a. Unhealthy diet: High sodium intake, low potassium intake, high saturated fat consumption (WHO, 2021)
- b. Physical inactivity (CDC, 2022)
- c. Overweight and obesity (BMI > 25 kg/m², particularly central obesity) (AHA, 2020)
- d. Tobacco use (smoking and smokeless tobacco) (WHO, 2021)
- e. Excessive alcohol consumption (CDC, 2022)
- f. Psychological stress and depression (AHA, 2020)

Clinical Features:

Many patients are asymptomatic. Common symptoms include:

- a. Headache (AHA, 2020)
- b. Dizziness (WHO, 2021)
- c. Fatigue (CDC, 2022)
- d. Blurred vision (AHA, 2020)
- e. Palpitation (WHO, 2021)
- f. Chest pain (CDC, 2022)
- g. Shortness of breath (AHA, 2020)
- h. Nose bleeding (epistaxis) (WHO, 2021)

Severe uncontrolled hypertension may lead to:

- a. Stroke (WHO, 2021)
- b. Heart attack (CDC, 2022)

- c. Kidney failure (AHA, 2020)
- d. Retinal damage (WHO, 2021)

4. Patient Profile

Particular	Patient Information
Name	Mr. Abdul Hamid
Age	52 years
Sex	Male
Religion	Muslim
Marital Status	Married
Occupation	Businessman
Address	Chattogram, Bangladesh
Educational Status	Illiterate
Family Members	5
Socioeconomic Status	Middle-income family
Date of Admission/Consultation	July 5, 2026
Medical Diagnosis	Hypertension

5. Chief Complaint

The patient complained of:

- a. Occasional headache for 2 weeks.
- b. Dizziness, especially after getting up suddenly.
- c. Occasional blurred vision.
- d. General weakness and fatigue.

Previous Medical History

The patient was diagnosed with hypertension approximately 2 years ago but did not always take his antihypertensive medication regularly. He has no known history of diabetes mellitus, tuberculosis, or major surgery. He has no known history of kidney disease or previous stroke.

Family History

The patient's father had a history of hypertension and died from a stroke at an older age. His mother had no known history of hypertension. No other significant hereditary disease was reported among the family members.

6. Diagnostic Investigation

Investigation	Normal Range	Patient's Result	Clinical Significance
Blood Pressure	<120/80 mmHg	160/100 mmHg	Significantly elevated; supports the diagnosis of hypertension.
Pulse Rate	60–100 beats/min	86 beats/min	Within normal range.
Respiratory Rate	12–20 breaths/min	18 breaths/min	Within normal range.
Hemoglobin	Male: 13–17 g/dL	14.2 g/dL	Within normal range.

Total WBC Count	4,000–11,000 cells/ μ L	7,800 cells/ μ L	Within normal range; no evidence of acute infection.
Serum Creatinine	0.7–1.3 mg/dL	1.0 mg/dL	Within normal range; indicates preserved renal function.
Serum Sodium	135–145 mmol/L	139 mmol/L	Within normal range.
Serum Potassium	3.5–5.0 mmol/L	4.2 mmol/L	Within normal range.
Fasting Blood Glucose	70–99 mg/dL	96 mg/dL	Within normal range.
Total Cholesterol	<200 mg/dL	218 mg/dL	Mildly elevated; increases cardiovascular risk.
ECG	Normal sinus rhythm	Normal sinus rhythm	No significant abnormality detected.

Patient's Result

The patient's blood pressure was 160/100 mmHg, which was significantly elevated. His total cholesterol was mildly increased at 218 mg/dL, while renal function, blood glucose, electrolytes, ECG, and other routine investigations were within normal limits.

Clinical Significance

The elevated blood pressure confirms poorly controlled hypertension and places the patient at increased risk of complications such as stroke, ischemic heart disease, heart failure, and kidney damage. The mildly elevated cholesterol further increases his cardiovascular risk. Normal creatinine and ECG findings indicate no obvious evidence of renal impairment or cardiac abnormality at the time of assessment.

7. Medical Diagnosis + Differential Diagnosis

Medical Diagnosis

Primary (Essential) Hypertension: The diagnosis is made based on the criteria of a systolic BP persistently ≥ 130 mmHg or diastolic BP persistently ≥ 80 mmHg on at least two separate occasions (ACC/AHA criteria), combined with a thorough history, physical examination, and basic investigations that rule out secondary causes. (ACC/AHA, 2017; WHO, 2021)

Differential Diagnosis (Ruling out Secondary Causes)

It is crucial to rule out secondary hypertension, especially in young patients, those with sudden onset, resistant hypertension, or specific clinical clues.

- a. Renal Parenchymal Disease: Ruled out by normal serum creatinine, eGFR, and absence of proteinuria/hematuria on urinalysis. (CDC, 2022)
- b. Renovascular Hypertension (Renal Artery Stenosis): Ruled out by lack of an abdominal bruit. (AHA, 2020)
- c. Primary Aldosteronism: Ruled out by a normal serum potassium level and absence of metabolic alkalosis. (WHO, 2021)
- d. Obstructive Sleep Apnea (OSA): Suspect in obese patients with snoring, daytime somnolence. (ACC/AHA, 2017)
- e. Coarctation of the Aorta: Ruled out by normal femoral pulses and equal blood pressure in upper and lower limbs. (CDC, 2022)
- f. Thyroid Disorders: Ruled out by lack of suggestive symptoms and a normal TSH. (AHA, 2020)

8. Medical Management, Pharmacological Management and Drug Adverse Effects

Medical Management (Non-Pharmacological Lifestyle Modifications)

This is the cornerstone of therapy for all patients.

- a. Dietary Sodium Reduction: Limit sodium intake to $< 2,300$ mg per day, ideally $< 1,500$ mg. (WHO, 2021)

- b. Dietary Approaches to Stop Hypertension (DASH) Diet: Rich in fruits, vegetables, whole grains, and low-fat dairy products. (AHA, 2020)
- c. Weight Reduction: Target a BMI <25 kg/m². A weight loss of 1 kg can reduce SBP by approximately 1 mmHg. (CDC, 2022)
- d. Regular Physical Activity: 150 minutes/week of moderate-intensity aerobic exercise. (WHO, 2021)
- e. Smoking Cessation: Complete cessation is mandatory. (AHA, 2020)
- f. Moderation of Alcohol Intake: Limit to ≤ 1 drink/day for women and ≤ 2 drinks/day for men. (CDC, 2022)

Pharmacological Management

- a. ACE Inhibitors (Example: Enalapril)
 - i. Action: Lowers blood pressure by relaxing blood vessels. (WHO, 2021)
 - ii. Adverse Effects: Dry cough, hypotension, hyperkalemia, dizziness. (AHA, 2020)
- b. Calcium Channel Blockers (Example: Amlodipine)
 - i. Action: Relaxes blood vessels and improves blood flow. (CDC, 2022)
 - ii. Adverse Effects: Ankle edema, headache, flushing, palpitation. (WHO, 2021)
- c. Diuretics (Example: Hydrochlorothiazide)
 - i. Action: Removes excess sodium and water. (AHA, 2020)
 - ii. Adverse Effects: Dehydration, hypokalemia, weakness, increased urination. (CDC, 2022)
- d. Beta Blockers (Example: Atenolol)
 - i. Action: Reduces heart rate and cardiac workload. (WHO, 2021)
 - ii. Adverse Effects: Bradycardia, fatigue, cold extremities, dizziness. (AHA, 2020)

9. Nursing Management

Assessment	Nursing Diagnosis	Planning	Nursing Interventions	Rationale	Evaluation
BP 160/100 mmHg with occasional headache and dizziness.	Ineffective health management related to inadequate adherence to antihypertensive treatment as evidenced by uncontrolled BP.	Patient will maintain BP within the recommended range and follow the treatment plan.	1. Monitor BP regularly. 2. Assess medication adherence. 3. Administer prescribed antihypertensive medication. 4. Encourage regular follow-up.	1. Monitors BP control. 2. Identifies reasons for poor adherence. 3. Helps control BP. 4. Promotes long-term management.	BP decreased and patient demonstrated better adherence to treatment.
Headache and dizziness with elevated BP.	Acute pain related to increased vascular pressure as evidenced by headache.	Patient will report reduced headache and dizziness.	1. Assess severity of headache and dizziness. 2. Provide a quiet, comfortable environment. 3. Encourage adequate rest. 4.	1. Determines severity and changes in symptoms. 2. Reduces environmental stimulation. 3. Promotes comfort. 4. Helps	Patient reported reduced headache and dizziness.

			Monitor BP and report severe elevation.	prevent complications.	
BP 160/100 mmHg and total cholesterol 218 mg/dL.	Risk for decreased cardiac output related to increased vascular resistance.	Patient will maintain stable cardiovascular status and vital signs.	1. Monitor BP, pulse, and cardiac symptoms. 2. Observe for chest pain, dyspnea, or palpitations. 3. Encourage low-salt and low-fat diet. 4. Encourage appropriate physical activity.	1. Detects cardiovascular changes early. 2. Helps identify complications promptly. 3. Helps reduce BP and cardiovascular risk. 4. Supports cardiovascular health.	Patient maintained stable vital signs with no signs of cardiovascular complications.
Patient is illiterate and does not take antihypertensive medication regularly.	Deficient knowledge related to inadequate information about hypertension and its management.	Patient will explain the importance of medication, diet, and follow-up.	1. Explain hypertension in simple language. 2. Teach the importance of taking medication regularly. 3. Advise	1. Improves understanding of the disease. 2. Prevents uncontrolled BP. 3. Reduces cardiovascular risk. 4.	Patient correctly explained medication, dietary restrictions, and BP-monitoring practices.

			reduction of salt and fatty foods. 4. Teach the importance of regular BP monitoring and follow-up.	Helps detect poor BP control early.	
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10. Health Education

Patients with hypertension should be educated about:

- a. Taking medications regularly (WHO, 2021)
- b. Reducing salt intake (AHA, 2020)
- c. Maintaining ideal body weight (CDC, 2022)
- d. Eating fruits and vegetables (WHO, 2021)
- e. Avoiding smoking and alcohol (AHA, 2020)
- f. Exercising regularly (CDC, 2022)
- g. Managing stress (WHO, 2021)
- h. Monitoring blood pressure regularly (AHA, 2020)
- i. Attending regular medical checkups (CDC, 2022)

Dietary approaches such as low-fat and low-salt diets are recommended. (WHO, 2021)

11. Disease Prognosis

The prognosis of hypertension depends on:

- a. Early diagnosis (WHO, 2021)
- b. Treatment adherence (AHA, 2020)
- c. Lifestyle modification (CDC, 2022)
- d. Control of risk factors (WHO, 2021)

Controlled hypertension has a good prognosis. However, uncontrolled hypertension may lead to severe complications such as:

- a. Stroke (WHO, 2021)
- b. Heart failure (AHA, 2020)
- c. Myocardial infarction (CDC, 2022)
- d. Kidney failure (WHO, 2021)
- e. Blindness (AHA, 2020)

Early intervention greatly improves quality of life and reduces mortality. (CDC, 2022)

12. Conclusion

Hypertension is a major global health problem affecting millions of people. It is a chronic condition that often remains unnoticed until complications develop. Proper diagnosis, effective treatment, healthy lifestyle practices, and patient education are essential for prevention and control. Nurses play an important role in assessment, monitoring, education, and promoting adherence to treatment. Community awareness and preventive measures can significantly reduce the burden of hypertension. (WHO, 2021; CDC, 2022)

13. References

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MALARIA CASE STUDY

1. Case Introduction

Malaria is a life-threatening parasitic disease caused by *Plasmodium* species and transmitted through the bite of infected female *Anopheles* mosquitoes. It remains one of the major public health problems worldwide, especially in tropical and subtropical countries. Malaria can affect people of all ages and may lead to severe complications if not treated promptly. (WHO, 2023; CDC, 2022)

2. Definition

Malaria is an acute febrile illness caused by infection of red blood cells by protozoan parasites of the genus *Plasmodium*, including *P. falciparum*, *P. vivax*, *P. ovale*, *P. malariae*, and *P. knowlesi*. It is defined clinically by the presence of fever or a history of fever within the past 48 hours (72 hours for *P. malariae*) along with laboratory confirmation of the parasite, and is classified as either uncomplicated or severe malaria. (WHO, 2023)

3. Epidemiological Study

Global and National Burden:

Global: According to the World Health Organization (WHO), there were an estimated 249 million malaria cases in 85 endemic countries in 2022, resulting in approximately 608,000 deaths. Children under five in sub-Saharan Africa bear the highest mortality burden. (WHO World Malaria Report, 2023)

National Situation (Bangladesh): Bangladesh has made remarkable progress, reducing confirmed cases by over 90% since the early 2000s. Approximately 90–95% of the total national case burden is concentrated in three hill tract districts: Rangamati, Bandarban, and Khagrachari. *P. falciparum* is the dominant species (>80% of infections). The National Malaria Elimination Programme (NMEP) aims to eliminate local transmission by 2030. (NMEP Bangladesh, 2022; Haque et al., 2020)

Mode of Transmission:

- a. Vector Bite: Primary mode is the bite of an infective female *Anopheles* mosquito (major vectors: *An. minimus*, *An. dirus*, *An. philippinensis*), typically between dusk and dawn. (WHO, 2023)
- b. Congenital Malaria: Rare, via transplacental transmission. (CDC, 2022)
- c. Blood-borne Transmission: Through transfusion of infected blood, contaminated needles, or organ transplantation. (WHO, 2023)

Risk Factors:

- a. Living in or traveling to endemic districts (especially Chittagong Hill Tracts). (NMEP, 2022)
- b. Outdoor occupations (forestry, "jum" farming). (Haque et al., 2020)
- c. Sleeping outdoors or without an insecticide-treated net (ITN). (WHO, 2023)
- d. Lack of access to healthcare. (CDC, 2022)
- e. High-risk groups: children under 5, pregnant women, immunocompromised individuals, non-immune travelers. (WHO, 2023)

Clinical Features:

- a. Uncomplicated Malaria: Classic paroxysm with three stages: cold stage (rigors 15–60 min), hot stage (high fever $>40^{\circ}\text{C}$, headache, vomiting 2–6 hours), sweating stage (profuse sweating, exhaustion). Splenomegaly and mild jaundice may be present. (WHO, 2023; Phillips et al., 2017)
- b. Severe Malaria: Caused mainly by *P. falciparum*; features include cerebral malaria (unrousable coma), severe anemia (Hb <5 g/dL), acute kidney injury (blackwater fever), ARDS, metabolic acidosis, hypoglycemia, shock, or parasite density $>5\%$. (WHO, 2023; CDC, 2022)

4. Patient Profile

Particular	Patient Information
Name	Mr. Nisang Marma
Age	28 years
Sex	Male
Religion	Buddhist
Marital Status	Married
Occupation	Farmer
Address	Rangamati, Bangladesh
Educational Status	Class 5
Family Members	7
Socioeconomic Status	Middle-income family
Date of Admission/Consultation	July 10, 2026
Medical Diagnosis	Malaria

5. Chief Complaint

The patient complained of:

- a. Intermittent high-grade fever for 4 days.
- b. Chills and shivering, especially during fever episodes.
- c. Severe sweating after fever.
- d. Headache and generalized body ache.
- e. Weakness and loss of appetite.
- f. Nausea and occasional vomiting.

Previous Medical History

The patient had no history of diabetes mellitus, hypertension, tuberculosis, or chronic liver or kidney disease. He had experienced an episode of malaria approximately 3 years ago, which was treated successfully. There was no history of recent surgery or hospitalization.

Family History

There was no significant family history of chronic disease. No other family member was currently suffering from malaria. However, the family lives in an area where mosquito exposure is common, particularly during the rainy season.

6. Diagnostic Investigation

Investigation	Normal Range	Patient's Result	Clinical Significance
Body Temperature	36.5–37.5°C	39.5°C	High fever, commonly associated with acute malaria.
Hemoglobin	Male: 13–17 g/dL	11.2 g/dL	Mild anemia, which may occur due to destruction of red blood cells.
Total WBC Count	4,000–11,000 cells/ μ L	4,200 cells/ μ L	Within normal range.

Platelet Count	150,000–450,000/ μ L	105,000/ μ L	Thrombocytopenia, which can occur in malaria.
Blood Glucose	70–140 mg/dL	92 mg/dL	Within normal range.
Serum Creatinine	0.7–1.3 mg/dL	1.1 mg/dL	Within normal range; no evidence of renal impairment.
Serum Bilirubin	0.2–1.2 mg/dL	1.8 mg/dL	Mildly elevated; may be associated with hemolysis.
Peripheral Blood Film	No malarial parasite	Malarial parasites detected	Confirms malaria infection.
Malaria Rapid Diagnostic Test	Negative	Positive for <i>Plasmodium</i>	Supports the diagnosis of malaria.

Patient's Result

The patient had a temperature of 39.5°C, mild anemia, reduced platelet count, and mildly elevated bilirubin. Peripheral blood film showed malarial parasites, and the malaria rapid diagnostic test was positive.

Clinical Significance

The investigation findings are consistent with acute malaria infection. The presence of malarial parasites confirms the diagnosis. Anemia may result from destruction of infected and uninfected red blood cells, while thrombocytopenia is a common hematological finding in malaria. The patient should be monitored closely for complications such as severe anemia, hypoglycemia, renal impairment, altered consciousness, and severe malaria.

7. Medical Diagnosis + Differential Diagnosis

Medical Diagnosis

- a. Microscopy (Gold Standard): Giemsa-stained thick and thin peripheral blood smears. Thick smear detects parasites and quantifies density; thin smear identifies species. (WHO, 2023)
- b. Rapid Diagnostic Tests (RDTs): Detect antigens (HRP2 for *P. falciparum*, pLDH for pan-malarial species); widely used in Bangladesh. (NMEP, 2022)
- c. Molecular Diagnosis (PCR): High sensitivity for low parasitemia and mixed infections. (Phillips et al., 2017)

Differential Diagnosis

- a. For Uncomplicated Malaria: Dengue fever, Chikungunya, Typhoid fever, Leptospirosis, Influenza, Viral hepatitis, Urinary tract infection. (WHO, 2023; CDC, 2022)
- b. For Severe Malaria: Sepsis, Meningitis/Encephalitis, Heat stroke, Complicated enteric fever, Acute viral hemorrhagic fevers. (Phillips et al., 2017)

8. Medical Management, Pharmacological Management and Drug Adverse Effects

Medical Management

Guided by species, parasite density, clinical severity, and patient factors. For severe malaria: airway protection, fluid balance, correction of hypoglycemia, seizure management, and possible renal replacement therapy. (WHO, 2023; NMEP, 2022)

Pharmacological Management (Based on Bangladesh National Malaria Treatment Guidelines)

- a. Uncomplicated *P. falciparum* Malaria:
 - i. Artemether (20 mg) + Lumefantrine (120 mg) (Coartem): 4 tablets twice daily for 3 days. (NMEP, 2022)
- b. Uncomplicated *P. vivax* Malaria:
 - i. Chloroquine (150 mg base): Day 1: 4 tabs; Day 2 & 3: 3 tabs each. (WHO, 2023)
 - ii. Primaquine (0.25 mg/kg daily for 14 days): Must check G6PD status first. (CDC, 2022)
- c. Severe Malaria:

- i. Injectable Artesunate (IV/IM): 2.4 mg/kg at 0, 12, 24 hours, then once daily until oral ACT tolerated. (WHO, 2023; NMEP, 2022)

Drug Adverse Effects

- a. Artemether-Lumefantrine: Nausea, headache, dizziness, palpitations. (Phillips et al., 2017)
- b. Artesunate (Injectable): Post-artesunate delayed hemolysis (rare). (WHO, 2023)
- c. Chloroquine: Pruritus (common in dark-skinned individuals), headache, blurred vision. (CDC, 2022)
- d. Primaquine: Acute hemolytic anemia in G6PD deficiency. (WHO, 2023)

9. Nursing Management

Assessment	Nursing Diagnosis	Planning	Nursing Interventions	Rationale	Evaluation
Temperature 39.5°C with chills and sweating.	Hyperthermia related to infectious process as evidenced by body temperature of 39.5°C.	Patient's temperature will return to normal range.	1. Monitor temperature regularly. 2. Provide light clothing and adequate ventilation. 3. Encourage fluids. 4. Administer	1. Monitors fever pattern. 2. Promotes heat loss. 3. Prevents dehydration. 4. Treats the infection and reduces	Temperature decreased and patient was more comfortable.

			prescribed antimalarial and antipyretic drugs.	fever.	
Headache, body ache, weakness, and fever.	Acute pain related to infectious process as evidenced by headache and body ache.	Patient will report reduced pain and improved comfort.	1. Assess pain intensity. 2. Provide rest in a quiet environment. 3. Encourage adequate fluids. 4. Administer prescribed medication.	1. Determines severity of pain. 2. Promotes comfort and rest. 3. Helps maintain hydration. 4. Relieves pain and treats the cause.	Patient reported decreased headache and body ache.
Fever, sweating, vomiting, and reduced fluid intake.	Risk for deficient fluid volume related to increased fluid loss	Patient will maintain adequate hydration and	1. Monitor intake and output. 2. Assess for dehydration. 3. Encourage	1. Assesses fluid balance. 2. Detects dehydration early. 3.	Patient remained adequately hydrated with stable vital signs.

	from fever, sweating, and vomiting.	normal urine output.	ORS and oral fluids. 4. Monitor vital signs.	Replaces fluid loss. 4. Detects changes caused by fluid loss.	
Weakness and loss of appetite; Hb 11.2 g/dL.	Imbalanced nutrition: less than body requirements related to decreased appetite and illness.	Patient will improve food intake and maintain adequate nutrition.	1. Assess appetite and food intake. 2. Provide small, frequent meals. 3. Encourage nutritious, easily digestible foods. 4. Monitor weight if required.	1. Identifies nutritional needs. 2. Improves food tolerance. 3. Provides energy for recovery. 4. Monitors nutritional status.	Patient tolerated meals and appetite improved.
Patient lives in an area with frequent mosquito exposure.	Deficient knowledge related to inadequate information	Patient will verbalize methods of	1. Teach use of mosquito nets. 2. Encourage	1. Reduces mosquito bites. 2. Provides	Patient correctly explained malaria prevention

	about malaria prevention.	preventing malaria.	use of mosquito repellents. 3. Teach elimination of stagnant water around the home. 4. Explain importance of completing antimalarial treatment.	additional protection. 3. Reduces mosquito breeding sites. 4. Promotes effective treatment and prevents complications.	and treatment measures.
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10. Health Education

- a. Use insecticide-treated mosquito nets while sleeping. (WHO, 2023)
- b. Wear long-sleeved clothing and use mosquito repellents. (CDC, 2022)
- c. Avoid stagnant water near home. (NMEP, 2022)
- d. Maintain environmental cleanliness. (WHO, 2023)
- e. Seek early medical treatment for fever. (CDC, 2022)
- f. Complete prescribed medications properly. (Phillips et al., 2017)

11. Disease Prognosis

The prognosis of malaria is good if diagnosed and treated early. Delayed treatment can lead to severe complications such as cerebral malaria, severe anemia, kidney failure, and death. Prevention and prompt treatment significantly reduce mortality. (WHO, 2023; Phillips et al., 2017)

12. Conclusion

Malaria is a major infectious disease caused by *Plasmodium* parasites and transmitted by mosquitoes. It continues to affect millions worldwide, particularly in tropical countries. Proper prevention, early diagnosis, effective treatment, and community awareness are important to reduce the burden of malaria. Nurses play a vital role in patient care, prevention, and health education. (WHO, 2023; CDC, 2022)

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ANXIETY DISORDER CASE STUDY

1. Case Introduction

Anxiety disorder is a common mental health condition characterized by excessive fear, worry, nervousness, and apprehension that interfere with daily activities. Anxiety can affect physical, emotional, cognitive, and social functioning. It is one of the most common psychiatric disorders worldwide and may occur in people of all ages. Early diagnosis and proper management are essential to improve quality of life and prevent complications. (WHO, 2023; APA, 2013)

2. Definition

Anxiety disorder is defined as a group of mental disorders characterized by clinically significant fear, anxiety, or avoidance that persists for at least several months and causes impairment in social, occupational, or other important areas of functioning. This includes generalized anxiety disorder (GAD), panic disorder, social anxiety disorder, and specific phobias. The DSM-5 criteria require that symptoms are not attributable to substance use or another medical condition. (APA, 2013; Stein & Sareen, 2015)

3. Epidemiological Study

Global and National Burden:

- a. Global: Anxiety disorders affect an estimated 301 million people worldwide (2019 data), with higher prevalence in women than men. Onset is typically in childhood or early adulthood. WHO ranks them as a leading cause of years lived with disability (YLD). (WHO, 2023; Bandelow et al., 2017)
- b. National Situation (Bangladesh): The 2018–19 national mental health survey reported a prevalence of any mental disorder among adults at 18.7%, with anxiety disorders contributing substantially. Prevalence is higher among women, urban slum dwellers, and those facing financial stress. The treatment gap exceeds 90% due to a shortage of psychiatrists (≈ 0.2 per 100,000 population). (NIMH Bangladesh, 2021; WHO, 2023)

Mode of Transmission:

Anxiety disorder is not transmissible. It develops from biological, psychological, and environmental factors. (APA, 2013)

Risk Factors:

- a. Biological: Female sex, family history, neurotransmitter imbalances (serotonin, norepinephrine, GABA). (Stein & Sareen, 2015)
- b. Psychological: Personality traits (neuroticism, behavioral inhibition). (Bandelow et al., 2017)
- c. Social/Environmental: Childhood adversity, trauma, chronic medical illness, substance abuse, low socioeconomic status, academic/occupational pressure. (WHO, 2023; APA, 2013)

Clinical Features (Generalized Anxiety Disorder):

- a. Psychological: Persistent worry, dread, hypervigilance, fear of losing control. (Stein & Sareen, 2015)
- b. Somatic: Palpitations, chest tightness, shortness of breath, dizziness, sweating, trembling, GI distress. In Bangladesh, patients often present with "whole-body pain" or "heart weakness." (NIMH Bangladesh, 2021)
- c. Behavioral: Avoidance, reassurance-seeking, agitation. (APA, 2013)
- d. Specific disorders: Panic attacks (abrupt intense fear), social anxiety disorder (fear of scrutiny). (WHO, 2023)

4. Patient Profile

Particular	Patient Information
Name	Mr. Salauddin
Age	24 years
Sex	Female
Religion	Muslim

Marital Status	Unmarried
Occupation	Unemployed
Address	Chattogram, Bangladesh
Educational Status	HSC
Family Members	5
Socioeconomic Status	Middle-income family
Date of Admission/Consultation	May 12, 2026
Medical Diagnosis	Anxiety

5. Chief Complaint

The patient complained of:

- Excessive worry and fear for 1 month.
- Difficulty sleeping for 2 weeks.
- Palpitation and increased heart rate during stressful situations.
- Restlessness and difficulty concentrating.
- Occasional headache and fatigue.

Previous Medical History

The patient had no history of hypertension, diabetes mellitus, thyroid disease, or other major chronic illnesses. She had no previous psychiatric hospitalization or diagnosed mental health disorder. She reported occasional episodes of worry in the past but had never received treatment for anxiety.

Family History

There was no significant family history of psychiatric illness. Her mother had hypertension, but there was no known family history of diabetes, neurological disease, or substance abuse.

6. Diagnostic Investigation

Investigation	Normal Range	Patient's Result	Clinical Significance
Blood Pressure	<120/80 mmHg	128/82 mmHg	Slightly elevated, possibly associated with anxiety.
Pulse Rate	60–100 beats/min	104 beats/min	Mild tachycardia, which may occur during anxiety.
Respiratory Rate	12–20 breaths/min	22 breaths/min	Mildly increased, possibly related to anxiety and stress.
Hemoglobin	Female: 12–16 g/dL	12.8 g/dL	Within normal range.
Total WBC Count	4,000–11,000 cells/ μ L	7,200 cells/ μ L	Within normal range; no evidence of acute infection.
Fasting Blood Glucose	70–99 mg/dL	91 mg/dL	Within normal range.
TSH	0.4–4.0 mIU/L	2.1 mIU/L	Within normal range; makes thyroid dysfunction less likely as a cause of symptoms.
ECG	Normal sinus	Normal sinus	No significant cardiac

	rhythm	rhythm	abnormality detected.
Anxiety assessment scale	Depends on scale	Moderate anxiety	Indicates clinically significant anxiety symptoms requiring appropriate management.

Patient's Result

The patient showed mild tachycardia and increased respiratory rate, while routine blood investigations, blood glucose, TSH, and ECG were within normal limits. Anxiety assessment indicated moderate anxiety.

Clinical Significance

The findings are consistent with anxiety, particularly because the patient has excessive worry, restlessness, sleep disturbance, palpitations, and difficulty concentrating. Normal TSH and ECG findings help rule out some physical conditions that can produce similar symptoms. The patient requires psychological support, stress-management education, and appropriate clinical follow-up.

7. Medical Diagnosis + Differential Diagnosis

Medical Diagnosis

- DSM-5 Criteria and structured tools like GAD-7 (score ≥ 10 indicates high probability). A validated Bangla version of GAD-7 is available in Bangladesh. (APA, 2013; NIMH Bangladesh, 2021)
- Diagnosis is clinical, based on history and physical examination after excluding medical causes. (Stein & Sareen, 2015)

Differential Diagnosis

- Medical Conditions: Hyperthyroidism, hypoglycemia, pheochromocytoma, cardiac arrhythmias, COPD. (WHO, 2023)

- b. Substance-Induced: Caffeine intoxication, alcohol/benzodiazepine withdrawal, amphetamine use. (APA, 2013)
- c. Other Psychiatric Disorders: Major depressive disorder, bipolar disorder, somatic symptom disorder. (Bandelow et al., 2017)

8. Medical Management, Pharmacological Management and Drug Adverse Effects

Medical Management (Non-Pharmacological)

- a. First-line: Cognitive Behavioral Therapy (CBT) – restructuring maladaptive thoughts and exposure therapy. In Bangladesh, available in tertiary hospitals and NGOs. (Stein & Sareen, 2015; NIMH Bangladesh, 2021)
- b. Counseling, mindfulness-based stress reduction, relaxation techniques (deep breathing, progressive muscle relaxation). (Bandelow et al., 2017)
- c. Lifestyle modification: regular exercise, sleep hygiene, reduced caffeine. (WHO, 2023)

Pharmacological Management

- a. SSRIs (First-line):
 - i. Escitalopram: 10–20 mg once daily. (Bandelow et al., 2017)
 - ii. Sertraline: 50–200 mg once daily. (Stein & Sareen, 2015)
 - iii. Start low, titrate slowly to avoid initial jitteriness. (APA, 2013)
- b. SNRIs:
 - i. Venlafaxine XR: 75–225 mg once daily. (WHO, 2023)
 - ii. Duloxetine: 60–120 mg once daily (good for anxiety with chronic pain). (Bandelow et al., 2017)
- c. Benzodiazepines (Short-term only, max 2–4 weeks):
 - i. Diazepam or Clonazepam for acute agitation/severe panic. (Stein & Sareen, 2015)
- d. Buspirone: Non-benzodiazepine anxiolytic, slow onset. (APA, 2013)

Drug Adverse Effects

- a. SSRIs/SNRIs: Nausea, headache, insomnia or initial increased anxiety, sexual dysfunction, weight gain. Discontinuation syndrome with abrupt stopping. (Bandelow et al., 2017; Stein & Sareen, 2015)
- b. Benzodiazepines: Drowsiness, dizziness, cognitive slowing, anterograde amnesia, dependence, withdrawal syndrome. (APA, 2013; WHO, 2023)

9. Nursing Management

Assessment	Nursing Diagnosis	Planning	Nursing Interventions	Rationale	Evaluation
Excessive worry, fear, restlessness, pulse 104 beats/min.	Anxiety related to perceived stress as evidenced by excessive worry, restlessness, and tachycardia.	Patient will demonstrate reduced anxiety and improved relaxation.	1. Assess anxiety level. 2. Provide a calm environment. 3. Encourage slow, deep breathing. 4. Stay with and reassure the patient.	1. Determines severity of anxiety. 2. Reduces external stressors. 3. Promotes relaxation. 4. Provides emotional support and security.	Patient appeared calmer and reported reduced anxiety.
Difficulty sleeping for 2 weeks and fatigue.	Disturbed sleep pattern related to anxiety as evidenced by difficulty sleeping and	Patient will report improved sleep and reduced fatigue.	1. Assess sleep pattern. 2. Encourage relaxation before bedtime. 3.	1. Identifies sleep problems. 2. Promotes relaxation. 3. Facilitates uninterrupted	Patient reported improved sleep and reduced fatigue.

	fatigue.		Reduce noise and disturbances at night. 4. Encourage a regular sleep routine.	sleep. 4. Helps establish healthy sleep habits.	
Difficulty concentrating and excessive worry.	Ineffective coping related to stress as evidenced by excessive worry and difficulty concentrating .	Patient will demonstrate effective coping strategies.	1. Encourage the patient to express feelings. 2. Identify sources of stress. 3. Teach simple stress-management techniques. 4. Encourage supportive family interaction.	1. Helps reduce emotional tension. 2. Identifies factors contributing to anxiety. 3. Improves coping ability. 4. Provides emotional support.	Patient expressed feelings and demonstrated appropriate coping methods.
Patient has limited knowledge about anxiety management.	Deficient knowledge related to lack of information about anxiety management	Patient will verbalize understanding of anxiety management .	1. Explain anxiety in simple terms. 2. Teach breathing and relaxation	1. Improves understanding of the condition. 2. Helps control anxiety symptoms. 3.	Patient correctly explained anxiety-management techniques

	.		techniques. 3. Encourage regular physical activity as tolerated. 4. Advise follow-up with a healthcare professional if symptoms persist or worsen.	Promotes mental well-being. 4. Ensures timely professional support.	and when to seek help.
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10. Health Education

- a. Normalize the condition: anxiety is a medical condition, not a character flaw. Physical symptoms are real, not a weak heart. (Stein & Sareen, 2015)
- b. Lifestyle as medicine: 30-minute daily brisk walking, avoid excessive tea/caffeine and social media scrolling. (Bandelow et al., 2017)
- c. Breathing techniques: box breathing (inhale 4s, hold 4s, exhale 4s, hold 4s) to stop panic attacks. (APA, 2013)
- d. Family psychoeducation: support rather than criticism; "I am here with you" is therapeutic. (NIMH Bangladesh, 2021)
- e. Treatment adherence: antidepressants take 2–4 weeks to work; complete 6–12 month course to prevent relapse. (WHO, 2023)

11. Disease Prognosis

The prognosis of anxiety disorders is chronic but manageable. Approximately 60–70% of patients achieve significant symptom remission with guideline-concordant treatment (CBT + SSRI). Untreated anxiety is a major risk factor for major depressive disorder, substance abuse, and adverse cardiovascular outcomes. In Bangladesh, poor drug adherence and relapse are driven by stigma and financial burden. (Bandelow et al., 2017; Stein & Sareen, 2015; NIMH Bangladesh, 2021)

12. Conclusion

Anxiety disorder is a common mental health problem that affects emotional and physical well-being. It can interfere with daily life if not properly managed. Early identification, psychological support, medication, and health education are essential for recovery. Nurses have an important role in providing emotional support, patient education, and mental health care. (WHO, 2023; APA, 2013)

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